# Computational Homework Assignments
# ASEN 6062: Celestial Mechanics
# Spring 2026

Please implement and solve the following computational homework problems using the language / tool of your choice. Details of your coding need not be shown, but can be requested and must be provided on demand. For each problem, please provide the specific information and plots requested. For numerical integrations, you are free to use whatever toolbox you need. Note that MATLAB is available to all students and can be used to implement all of these problems.
Computational HW Problems #1 & 2: Due March 13, 2026.
Computational HW Problem #3: Due April 3, 2026.

1. Write a simulation program to compute the dynamical motion in the 3-body problem. To do this use the equations of motion expressed in the Jacobi Coordinates introduced in class, and use a dimensionless system defined so that the masses sum to unity ($m_1+m_2+m_3 = 1$) and with the gravitational constant taken as unity ($\mathcal{G} = 1$). Your program must be able to take in system parameters (i.e., masses) and initial conditions (i.e., relative positions and relative velocities) and produce a time series of positions and velocities for the dynamical evolution of the system over an arbitrary time. Note that for the given units if you choose your relative positions to be of order unity, then the velocities for bounded motion should also be of order unity.

   (a) For a set of unequal masses and "arbitrary" initial relative positions and velocities, show that your simulation conserves energy $E$ and angular momentum $H$ at a level consistent with the numerical precision of your approach. To show this, plot the time history of these quantities and show their relative variation from their initial values. You will generally see some small variation due to numerical integration errors – to verify this repeat the same calculation but at a higher numerical tolerance and verify that the small variations consistently decrease in magnitude.

   (b) For the initial masses 1/2, 1/3, 1/6 place the system into an initial relative equilibrium Lagrange solution, i.e., place them at the vertices of an equilateral triangle with unit distance between them and give them all a common rotation rate about the center of mass of unity. The time history of this system should remain consistent for a while, but as this is an unstable set of masses it will eventually diverge. Repeat, giving a small shift in the initial location of one of the bodies (on the order of 0.001). Now how rapidly does the system depart from its relative equilibrium? Show and compare the resultant trajectories.

   (c) Repeat the above for initial masses that satisfy the Routh Stability Criterion $m_1m_2 + m_2m_3 + m_3m_1 < 1/27$. Verify that the system now stays close to its initial configuration when given a small shift in its initial conditions. How far can you shift the initial relative positions before it changes significantly from its triangular configuration?

2. Make a copy of the previous program and modify it to simulate motion in the 5-body problem. Again use the relevant Jacobi Coordinate form of the equations of motion, choose the total mass such that $m_1 + m_2 + m_3 + m_4 + m_5 = 1$, etc.

   Note, if you wish you can also use the on-line tool REBOUND for this simulation: https://rebound.hanno-rein.de

   If you use REBOUND, please note this explicitly and also show a verification between your previous 3BP simulation and a REBOUND simulation.

   (a) Verify that your new system also conserves energy and angular momentum, similar to before.

   (b) For an equal mass system, start it in an initially circular orbit for the $N$-gon central configuration from the HW, making the distance from each body to the barycenter unity. This system should be unstable – comment on how fast it departs from this configuration.

   (c) Repeat the above, but now place the initial system on a mutually eccentric orbit with eccentricity $e = 0.1$. Note, it will be easiest to start such a system at periapsis or apoapsis just by giving the configuration a non-circular spin rate. Comment on how this system evolves as compared to the previous case.

   (d) Experiment with this problem to make a system where a 2-mass and a 3-mass system start in orbit about each other and then flyby each other. Describe how the system evolves – test a number of different initial conditions to sample some of the diversity of outcomes that can occur.

3. Create a "restricted 4-body" problem using your 3-body problem from Computational Homework Problem 1. To do this, you just need to add one more body that is affected by the other three bodies with mass, which however does not affect the other bodies – thus you can reuse your code from Problem 1 for the motion of the three massive bodies.

   To set up the problem, introduce a 4th body to the Jacobi coordinates of the 3-body problem, which will be located with a position $\mathbf{R}_3$ with location measured from the center of mass of the 3-body system. You will now have to also add the gravitational attraction of the three bodies on this massless fourth body. The motion of this new body will have to be computed along with the bodies from Problem 1.

   In the following, you will repeat your numerical integrations from parts a, b and c, but will now focus on the motion of the point mass driven by these bodies with mass. For each of the problems, start the infinitesimal mass at the initial center of mass of the three bodies (an initial position of $\mathbf{R}_3 = \mathbf{0}$ in the Jacobi coordinate notation), and plot its trajectory over time. For parts b and c, also compute the trajectory for the initially unperturbed Lagrangian solution and for the perturbed Lagrangian solution.

   In your HW answer, provide your equations of motion for body 4 and plot its trajectory for all of the requested cases. Describe the results that you find.